Progress Report 2

1. **Project title:**
   Swarm of robots to complete locomotive and object-searching tasks, and touchscreen interface for a multipurpose robot.

2. **Names, student numbers, and emails**:
   Group 36 members: (all members are registered in ECSE 458-D1)

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3. **Project advisor(s):**
   Joseph Vybihal: joseph.vybihal@mcgill.ca

4. **Group meetings and meetings with advisor(s):**
   All four team members, along with the advisor, have agreed to hold two meetings each week: one with all team members present and another with the advisor and any available team members at that time.

   Weekly meeting with the entire group: Tuesdays at 4pm (previously it was Thursdays)
   Weekly meeting with the advisor: Fridays at 2pm

   Outlined below is a detailed list of the meetings that have been attended with the advisor, including dates and any key discussions or decisions made during each session.
   Group meeting(s)
   - Thursday, 24th October
     - Filled out the logistical information required by project advisor for getting access to the code repos, etc
     - Contacted other groups involved to get access to the data available on the robots
     - Decided to present figma of the user interface and a control flow diagram to the project advisor for the midterm presentation
   - Tuesday, 29th October
     - Discussed the pros and cons of a doing website Vs an application for the user interface before deciding on the idea of a website
     - Decided on using an android tablet for the hardware
     - Discussed the details of how the interface will look, what information can be entered in it etc

- Tuesday, 5th November
    - Finalized the control flow diagram
    - Discussed tasks and roles for the presentation

Meeting(s) with Advisor

- Friday, 25th November
    - Met with the other capstone groups to understand their progress
- Thursday, 7th November
    - Did the presentation with the Figma design of the interface and the control flow diagram
    - Discussed our decision to do a website
    - We were advised to think about the decision more critically and the problems of having more overload on the robot
    - Discussed issues of connecting to the robot and give command to it
- Friday, 8th November
    - Met with the other capstone groups to understand their progress

Recent progress

Since the last progress report, we had a midterm presentation with our project advisor which essentially entailed us presenting our work and plans so far. We are focusing on building the user interface for the R1D4 robot for this semester.

Our team engaged in a comprehensive discussion to evaluate the pros and cons of developing the user interface as either a website or an application. This analysis led us to conclude that building a website would be the most efficient and flexible solution.

However, following feedback from our advisor during the midterm presentation, we are reconsidering our initial decision. We are now inclined with developing an application, as it offers the advantage of offline functionality. This feature would ensure that the interface remains operational even in the event of a network failure, and being more reliable.

The last few weeks we prepared for the presentation. We created a Figma design (can be viewed here: figma-design-for-r1d4) for the user interface which describes the features and functions we want to implement. The interface will have three main options:

Manage Tasks: show all scheduled deliveries with options to delete and modify these tasks.

Send Delivery: allows user to schedule a delivery to a specific location by adding all the details

Confirm Pickup: This page will show all the deliveries in a specific location and allow the user to view the item and confirm the pickup of that item.

In addition to these functions, the robot will display its emotions as well, for example, a thinking face when it is computing or a happy face when it successfully completes a task.

A control flow diagram was also created which shows the logical flow of the functioning of the robot with the interface.

A state diagram for the tasks was also created which provides a visual guide to task progression, enabling more effective development, tracking, and troubleshooting within the system.

Future plans

Our future plans for the capstone project are as follows. Our first next step will be accessing the ESP of the robot to gather essential data about its current operational state. This functionality is critical to ensure that the information displayed on the tablet is accurate and relevant, helping users monitor and interact with the robot seamlessly.

To accomplish this, we plan to develop either a socket program running on the ESP or have the ESP start up a C program upon a post request from the website.. By November 15, we intend to finalize the tablet hardware interface and launch initial work on this website based on the control diagram and figma design previously created. Following this, we'll implement the API (either socket or C program with POST request) that facilitates data transfer from the robot to the website, which will be designed to dynamically update based on the robot's status. By the end of November, our goal is to have the website fully operational, displaying the robot's current state accurately. We'll then mount the tablet on the robot itself, transforming it into an intuitive, user-friendly interface. Our final presentation to Prof. Vybihal, planned before December 4, will showcase the completed system.

Professionalism

For our project, keeping the public interest in mind is important since the final product will be working in a highly public setting, I.E the hospital. Throughout development, we'll focus on thorough testing, especially for the data flow between the ESP and the Android tablet, to prevent any miscommunication or system failures. System failures could cause a delay in a crucial medication delivery, so we must do our best to prevent this. In terms of the physical integration, we'll ensure the mounting of the Android tablet on the robot is secure, protecting users from any physical risks and preventing damage to the equipment. Although only adults will be interacting with this device, it will be in the public space so it will be best to make it as child proof as possible. By adhering to best practices in system design, we aim to not only meet but exceed the expectations of reliability and safety for engineering projects.